Arghan Dutta

Curriculum Vitae

Indian Institute of Science
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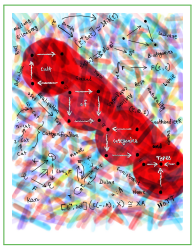
in *Arghan Dutta*



Research Interests

My research interests lies in the field of Categorical Homotopy Theory, Higher Category Theory and Homotopy Type Theory. I am particularly interested in the development of model independent theory of (∞, n) categories. I am also interested in the theory of (enriched) Model categories. I am the co-organizer of the Cult of Categories and Types (CoCT) seminars at Indian Institute of Science, Bangalore, India.

Education

2021–Ongoing **MS in Mathematics**, Indian Institute of Science, Bangalore, India
- Score : CGPA : 8.2/10.0

2023 **Master's Project**, *Simplicial Homotopy Type Theory for synthetic* $(\infty, 1)$*-categories*, Indian Institute of Science, Bangalore, India
- Under the supervision of Siddhartha Gadgil

2017–2020 **B.Sc. in Mathematics**, *Honors*, The Bhawanipur Education Society College, West Bengal, India
- Score : 87.375% / CGPA : 9.2/10.0

2017 **Central Board of Secondary Education**, *Higher Secondary*, Hariyana Vidya Mandir, West Bengal, India
- Score : 95% in Mathematics

2015 **Indian Certificate of Secondary Education**, *Secondary*, The Aryans School, West Bengal, India
- Score : 93% in Mathematics

Academic Achievements

2022 **Selected for the prestigious Ph.D. scholarship funded by NBHM-DAE**
- Organized by the National Board of Higher Mathematics, India

2021 **Selected for the Integrated MS-Ph.D. program**, *Indian Institute of Science (IISc)*
- Organized by the Indian Institute of Science, Bangalore, India

2021 **Secured All India Rank 33 in IIT-JAM Mathematics**, *Entrance Examination*
- Organized by the Indian Institute of Science, Bangalore, India

2021 **Cleared Tata Institute of Fundamental Research**, *Entrance Examination*
- Organized by the Tata Institute of Fundamental Research, Mumbai, India

2020 **Secured All India Rank 18 in MMath ISI, Kolkata**, *Entrance Examination*
- Organized by the Indian Statistical Institute, Kolkata, India

2020 **Cleared Chennai Mathematical Institute**, *Entrance Examination*
- Organized by the Chennai Mathematical Institute, Chennai, India

2019 **Selected for Mathematics Training and Talent Search**, *MTTS program*
- Organized by the Mathematics Training and Talent Search Trust, India

2017 **Cleared Joint Entrance Examination**, *Advanced*
- Organized by the Indian Institute of Technology Madras, India

2017 **Cleared Joint Entrance Examination**, *Mains*
- Organized by the National Testing Agency, India

Research

2023–Ongoing **Synthetic formalization of $(\infty, 1)$ Bordism categories**
- Master's thesis under the supervision of Siddhartha Gadgil

2023–Ongoing **Enriched cofibrantly generated Model Categories and Enriched Bousfield localizations of Model Categories**
- Under the supervision of Rekha Santhanam

Projects and Readings

2023 **Simplicial Homotopy Type Theory**, *Model independent $(\infty, 1)$ Category Theory*
- Master's project under the supervision of Siddhartha Gadgil

2022 **On the proof of Every Regular Local Ring is an UFD**
- Read under the supervision of Amalendu Krishna

2022 **On the Existence and Uniqueness of Haar Measure on Locally Compact Topological Groups**
- Course project under the supervision of Radhika Ganapathy

Workshops and Conferences attended

Jul–Aug 2023 **Applied Category Theory**, *6th International Conference*, Attended online, University of Maryland, College Park, Maryland, United States

May 2023 **Dualities in Topology and Geometry**, *Workshop*, International Centre for Theoretical Sciences, Tata Institute of Fundamental Research, India

Talks and Presentations

Sep 2023 **Strict 2-categories and Bicategories**, *Cult of Categories and Types*, CoCT Seminars, Indian Institute of Science, India

Sep 2023 **Structures on Monoidal Categories**, *Cult of Categories and Types*, CoCT Seminars, Indian Institute of Science, India

Sep 2023 **Small object argument and Cofibrantly generated Model Categories**, Online talk, Indian Institute of Technology, Bombay

Jun 2023 **Simpicial Homotopy Type Theory**, *Master's project presentation*, Indian Institute of Science, India

Dec 2022 **On the properties of maps between Zariski spaces induced by integral extensions**, Indian Institute of Science, India

Dec 2022 **On Positive Kernels and Tensor Product of Hilbert spaces**, Indian Institute of Science, India

Oct 2022 **On the Introduction to Categorical Logic**, *On the celebration of National Mathematics Day 2022*, Bhawanipur Education Society College, Kolkata

Oct 2022 **On the Introduction to Category Theory**, *Prime time seminars*, Indian Institute of Science, India

Linguistic Skills

Languages

- English
- Bengali

Programming Skills

- C
- C++
- Java
- Lean